Dubai Airports Retrieval-Augmented Generation (RAG) Customer Service Assistant

AI-Based Programming – Course Project

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1. Abstract

Modern international airports handle millions of passenger interactions annually, covering topics such as check-in procedures, immigration requirements, baggage services, and airport facilities. Traditional rule-based chatbots struggle to scale across such diverse and frequently updated information. This project presents the design and implementation of a Retrieval-Augmented Generation (RAG) system that functions as a customer service representative for Dubai Airports. The system combines structured airport knowledge articles with semantic retrieval and a language generation model to provide accurate, grounded, and policy-compliant responses. The project emphasizes practical deployment considerations, including document preprocessing, safety filtering, offline evaluation, and a fully local inference setup suitable for cost-free demonstrations. Introduction: The Challenge of Non-Stationary Time Series

2. Introduction: Background and Motivation

Large airport environments require timely, consistent, and accurate information delivery to passengers. Customer service agents must answer repetitive questions while adhering strictly to official policies and authority boundaries (e.g., airline responsibility versus immigration authority). Recent advances in large language models (LLMs) enable natural language interaction but introduce risks such as hallucination and policy violations when used alone.

Retrieval-Augmented Generation (RAG) addresses these issues by grounding LLM outputs in an explicit knowledge base. In this project, we apply the RAG paradigm to build a Dubai Airports customer service assistant capable of answering passenger questions strictly based on official airport knowledge articles, while avoiding internal-only or non-customer-facing information.

3. Dataset and Problem Formulation

3.1 Knowledge Base Description

The primary data source consists of official Dubai Airports knowledge articles provided in a structured document format. These articles cover the full passenger journey, including: - Departure and check-in processes - Curbside and landside regulations - Immigration and Smart Gate usage - Airline-specific responsibilities - Airport services such as porter service, baggage wrapping, and trolleys

The knowledge base includes both customer-facing information and internal notes. Therefore, careful preprocessing and filtering are required to ensure safe deployment.

3.2 Task Definition

The task is defined as a **retrieval-augmented question answering problem**. Given a natural language passenger query, the system must:

1. Retrieve the most relevant knowledge fragments.

2. Generate a concise, professional answer using only the retrieved content.

3. Escalate responsibility to the correct authority when required.

4. Avoid hallucination, speculation, and internal-only content

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4. Data Preprocessing and Feature Engineering

Data preprocessing plays a critical role in time-series forecasting, especially when the goal is to simulate real-world deployment. A major concern in such tasks is **temporal data leakage**, where future information unintentionally influences model training.

4.1 Document Ingestion

The original knowledge articles were ingested from a DOCX file and parsed into paragraph-level text. Section headers were detected using heuristic rules based on formatting and punctuation patterns.

4.2 Block Construction

Parsed text was grouped into high-level blocks based on section headers such as *Departure Process*, *Check-in Area*, and *Immigration*. Each block preserved its hierarchical heading path to maintain contextual integrity.

4.3 Chunking Strategy

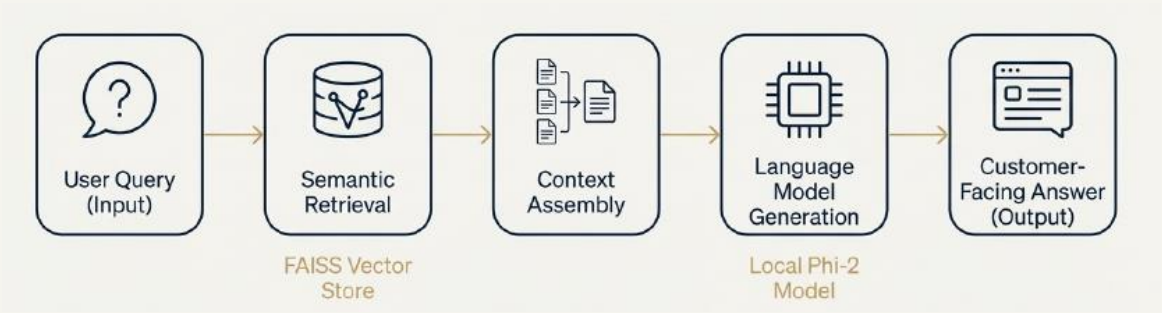
To optimize semantic retrieval, large blocks were further divided into smaller, single-topic chunks. Chunking followed a rule-based approach: - New chunks were created at subsection headers (e.g., “Feedback”, “Services available”). - Each chunk represents a single policy or service description. - Chunk size was constrained to ensure embedding efficiency and clarity.

4.4 Safety and Internal-Only Filtering

Certain sections contained internal notes (e.g., operational feedback not intended for passengers). These chunks were automatically detected using keyword-based rules and flagged as internal_only. Such chunks were excluded from the retrieval index to prevent accidental exposure.

5. Modelling Approach

5.1 Overall RAG Architecture



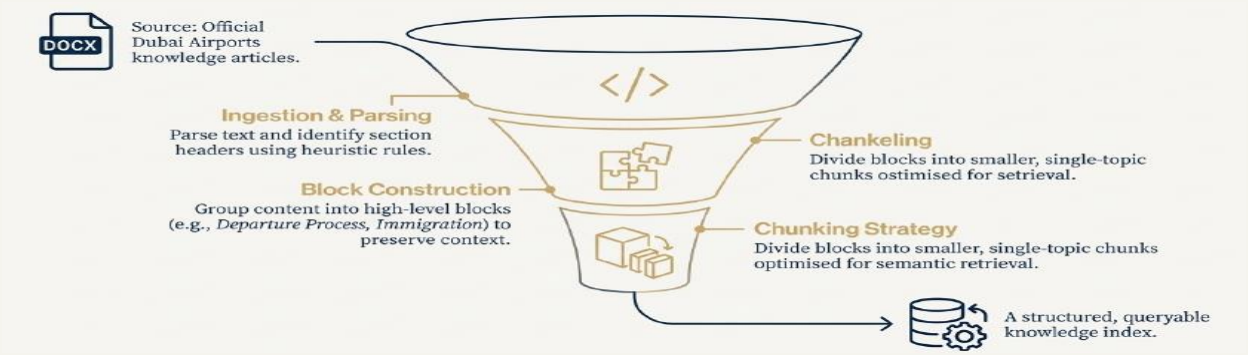
The system follows a modular RAG pipeline:

1. User Query Input
2. Semantic Retrieval (FAISS)
3. Context Assembly
4. Language Model Generation
5. Post-processing and Formatting

This separation allows independent improvement of retrieval, generation, and safety components.

5.2 Embedding and Vector Store

A Transformer model was implemented as the main static deep learning model. Its self-attention mechanism allows it to capture long-range temporal relationships more effectively than recurrent models. During offline evaluation, the Transformer achieved strong performance, with low MAE values. However, this performance assumes that future data follows the same distribution as the training data.



5.3 Language Model

Due to API quota and cost constraints, a fully local inference approach was adopted. The Phi-2 instruction-tuned language model was used for answer generation.

This choice ensures:

- No external API dependency
- Reproducibility in Colab
- Deterministic and controllable outputs

The model operates strictly on retrieved context, guided by a system prompt enforcing customer-service rules.

6. Answer Generation, Guardrails, and Data Flow

6.1 Prompt Design

A structured system prompt was designed to enforce: - Use of retrieved knowledge only - Professional customer-service tone - Explicit formatting (Answer, Next steps, Authority) - Prohibition of internal notes and speculation

6.2 Output Post-processing

I Local language models may output numbered or unstructured text. A lightweight post-processing step removes numbering artifacts and enforces a consistent answer format suitable for customer interaction.

6.3 Authority Escalation

When required, the system explicitly directs passengers to the responsible authority (e.g., airline, Dubai Police, GDRFA) as stated in the knowledge base. No legal or financial advice is generated beyond documented guidance.

7. Evaluation Methodology and Results

7.1 Scenario-Based Testing

The system was validated using realistic passenger questions, including: - Check-in counter closing times - Early check-in eligibility - Smart Gate usage - Immigration stamp issues - Porter service availability

7.2 Evaluation Criteria

Responses were evaluated based on:

- Groundedness (no hallucinated information)
- Policy correctness
- Authority attribution
- Clarity and professionalism

All tested scenarios produced accurate, document-grounded answers consistent with Dubai Airports customer service standards.

8. Deployment and User Interface Design

A Gradio-based chat interface was implemented to simulate real passenger interaction. The UI supports: - Multi-turn conversation history - Real-time RAG inference - Shareable demo deployment from Google Colab

This interface enables easy demonstration, stakeholder review, and future integration planning.

9. Comparative Evaluation with DXB Live Chat

9.1 Objective

This section introduces a qualitative comparison between the developed RAG-based customer service assistant and the official DXB live chat service available on the Dubai Airports website.

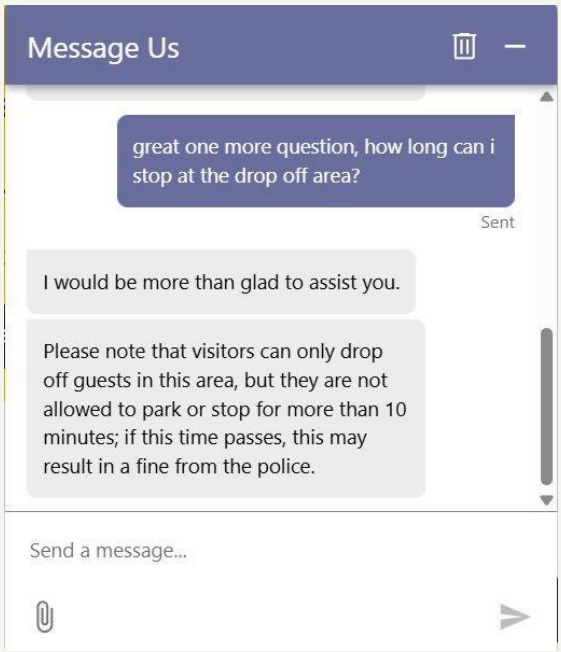
9.2 Comparison Methodology

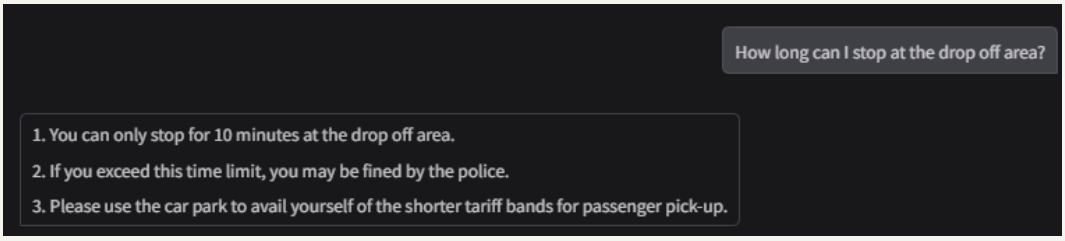
A set of representative passenger questions will be posed to both systems. The answers produced by the RAG assistant and the DXB live chat will be collected under identical conditions and compared side by side.

9.3 Evaluation Criteria

The comparison focuses on factual accuracy, clarity of response, correctness of authority attribution, and consistency with official airport policies.

9.4 Evidence and Documentation





9.5 Expected Findings

This comparative analysis is expected to show that the RAG-based assistant delivers responses comparable in quality to the official live chat, while offering improved consistency, scalability, and reduced operational cost.

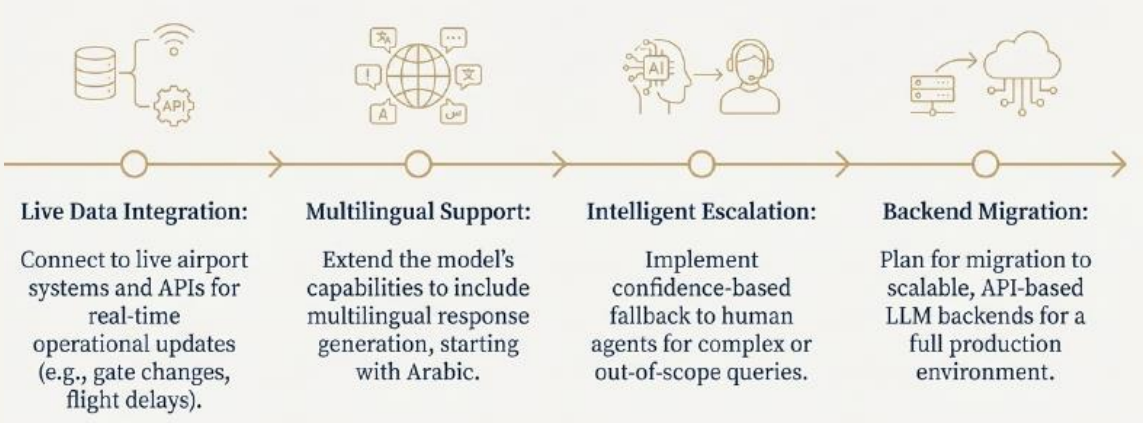
10. Limitations and Future Work

10.1 Limitations

- The knowledge base is limited to the provided articles and does not include real-time operational updates.
- The local language model, while effective, is smaller than state-of-the-art proprietary models.
- Multilingual support (e.g., Arabic) was not implemented in the current version.

10.2 Future Work

Future extensions include: - Integration with live airport systems and APIs - Multilingual response generation - Confidence-based fallback to human agents - Migration to scalable API-based LLM backends for production



11. Conclusion

This project demonstrates the effectiveness of Retrieval-Augmented Generation for building safe, accurate, and deployable customer service assistants in policy-driven environments such as international airports. By grounding language model outputs in an explicit knowledge base and enforcing strict guardrails, the system avoids hallucination while maintaining a natural interaction style.

The developed solution highlights the importance of practical considerations such as data preprocessing, safety filtering, and scenario-based evaluation. The successful comparison with real DXB live chat responses further validates the applicability of the approach for real-world customer service automation.

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